

Day 18 : REST API Integration

1. Introduction to REST API

A REST API allows your Flutter app to communicate with a server over the internet. APIs are used to fetch, send, update, and delete data from remote databases or services.

Examples:

- Login/Register systems
- News applications
- Weather applications
- E-commerce applications.

2. Making HTTP Request

Flutter uses packages like:

- **http** → Simple and beginner-friendly.
- **dio** → Advanced package with interceptors, file upload, timeout support, etc.

Add Dependency : http: ^1.2.0

3. HTTP Methods

A. GET Request

Used to fetch data.

```
final response = await http.get(
  Uri.parse('https://jsonplaceholder.typicode.com/posts'),
);
```

B. POST Request

Used to send new data.

```
final response = await http.post(
  Uri.parse('https://api.com/posts'),
  body: {
    'title': 'Flutter',
    'body': 'Learning API'
  },
);
```

C . PUT/PATCH Request

Used to update existing data

```
await http.put(  
  Uri.parse('https://api.com/posts/1'),  
  body: {  
    'title': 'Updated Title'  
  },  
);
```

D. DELETE Request

Used to remove data.

```
await http.delete(  
  Uri.parse('https://api.com/posts/1'),  
);
```

4. Headers and Query Parameters

Headers

Used to send extra information.

```
headers: {  
  'Content-Type': 'application/json',  
  'Authorization': 'Bearer token'  
}
```

Query Parameters

Example URL:

<https://api.com/users?page=1&limit=10>

```
Uri uri = Uri.https(  
  'api.com',  
  '/users',  
  {'page': '1', 'limit': '10'},  
);
```

5. Handling JSON Data

In pubspec.yaml:

```
dependencies:  
  supabase_flutter: ^2.0.0
```

Most APIs return data in **JSON format**.

Example JSON:

```
{  
  "id": 1,  
  "name": "Sneha",  
  "email": "sneha@gmail.com"  
}
```

Parse JSON Response

```
import 'dart:convert';

Map<String, dynamic> data =
    jsonDecode(response.body);

print(data['name']);
```

6. Creating Model Classes

Model classes convert JSON into Dart objects

```
class User {
    final int id;
    final String name;
    final String email;

    User({
        required this.id,
        required this.name,
        required this.email,
    });

    factory User.fromJson(Map<String, dynamic> json) {
        return User(
            id: json['id'],
            name: json['name'],
            email: json['email'],
        );
    }

    Map<String, dynamic> toJson() {
        return {
            'id': id,
            'name': name,
            'email': email,
        };
    }
}
```

fromJson()

Converts JSON → Dart Object.

```
User user = User.fromJson(jsonData);
```

toJson()

Converts Dart Object → JSON.

```
Map<String, dynamic> json = user.toJson();
```

7. Handle Nested JSON-

```
{
  "name": "Sneha",
  "address": {
    "city": "Pune",
    "country": "India"
  }
}
```

8 . Error Handling

Error handling prevents app crashes.

Try-Catch

```
try {
  final response = await http.get(url);
} catch (e) {
  print(e);
}
```

HTTP Status Codes

Code	Meaning
200	Success
201	Created
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Server Error

```
if (response.statusCode == 200) {
  print("Success");
} else {
  print("Error");
}
```

9. Network Errors

Occurs when:

- No internet
- Server down
- Wrong URL

1. Timeout Handling

```
final response = await http
    .get(url)
    .timeout(Duration(seconds: 10));
```

10. Loading States

Loading states improve user experience.

Show Loading Indicator

```
bool isLoading = true;

</> dart

isLoading
? CircularProgressIndicator()
: ListView()
```

Getting API Key :

Registration complete

Your API key is: `8a24150527d047cda99826321c6ad863`

For help getting started please look at our [getting started guide](#).

We post API status updates and other news on our Twitter feed, so please follow us there if that's important to you:

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